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Arboricultural Impact Assessment

Junction of Old Vicarage Lane and A420

Swindon

January 2023

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Rev	Issue Status	Prepared / Date	Approved/Date
-	Draft	MHE / 22.08.22	HCK / 24.08.22
A	Layout Update: 1335-HL-058-107F	MHE / 10.01.23	HCK / 10.01.23
B	TRP & TPP Viewport Scale Update	MHE / 20.01.23	HCK / 20.01.23

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1.0 INTRODUCTION

1.1 This report has been prepared by FPCR Environment and Design Limited on behalf of Hannick, Hallam and Taylor Wimpey to provide an Arboricultural Impact Assessment and details of those trees and hedgerows which would require removal and/or pruning to facilitate the proposed junction of Old Vicarage Lane and A420, South Marston, East of Swindon (hereafter referred to as 'the site').

1.2 The site is centered on Ordnance Survey grid reference SU 195 872.

1.3 The survey was carried out on 20th January 2022.

Scope of Assessment

1.4 The tree survey and assessment of existing trees has been carried out in accordance with guidance contained within British Standard 5837:2012 *'Trees in Relation to Design, Demolition and Construction - Recommendations'* (hereafter referred to as BS5837). The guidelines set out a structured assessment methodology to assist in determining which trees would be deemed either as being suitable or unsuitable for retention.

1.5 The guidance also provides recommendations for considering the relationship between existing trees and how those trees may integrate into designs for development; demolition operations and future construction processes so that a harmonious and sustainable relationship between any retained trees and built structures can be achieved.

1.6 The purpose of the report is therefore to firstly, present the results of an assessment of the existing trees' arboricultural value, based on their current condition and quality and to secondly, provide an assessment of impact arising from the proposed development of the site.

Planning History

1.7 The site has received planning approval (reference Application Number S/OUT/13/1555) for a large residential development on Land North of A420, South Marston and Rowborough.

1.8 The proposed development includes up to 2380 residential dwellings, mixed use local center/areas (including A1 retail up to 1,500m^2, services (A2), restaurants, pubs and takeaways (A3, A4, A5) business uses (B1) up to 1,000m^2, residential), community uses (D1), sheltered and/or care accommodation (C2, C3), two primary schools, green infrastructure including formal (including playing fields) and informal open space, landscape, biodiversity and amenity space (including LEPS, NEAPS/MUGA), changing and sports facilities (including D2 and ancillary retail uses), sustainable drainage systems; highway, cycle and pedestrian routes; car parking; infrastructure (including utilities); engineering works including ground remodeling; demolition, site reclamation and removal of structures; the formation of new accesses from the A420, Old Vicarage Lane and Thornhill Road, highway, cycle and pedestrian routes.

Purpose of the Summary

1.9 As part of the conditional outline approval, detailed designs for the points of access to the development are required and have been considered by this Arboricultural Assessment. There

will be three main points of access to the wider development and this Impact Assessment relates to the proposed Junction of Old Vicarage Lane and the A420.

1.10 The details provided within this Arboricultural Impact Assessment are also required to satisfy two tree related planning conditions of the Outline Planning Application, namely C12 and C13.

2.0 PLANNING POLICY

National Planning Policy Framework July 2021

2.1 National Planning Policy is defined by the National Planning Policy Framework (NPPF). This sets out the Government's most current and up to date planning policies for England and how these should be applied. The current NPPF is dated July 2021.

2.2 Paragraphs 10 and 11 of the NPPF state that there is a presumption in favour of sustainable development and states that for decision making, the LPA should be '*c) approving development proposals that accord with an up-to-date development plan without delay'.* In the absence of a development plan or the development plan is out of date, the acting LPA should grant planning consent so far as the development proposals do not breach the policies and guidance outlined in the NPPF.

2.3 In relation to arboriculture, the NPPF also states that:

- *131 'Trees make an important contribution to the character and quality of urban environments, and can also help mitigate and adapt to climate change. Planning policies and decisions should ensure that new streets are tree-lined (footnote 50), that opportunities are taken to incorporate trees elsewhere in developments (such as parks and community orchards), that appropriate measures are in place to secure the long-term maintenance of newly-planted trees, and that existing trees are retained wherever possible. Applicants and local planning authorities should work with highways officers and tree officers to ensure that the right trees are planted in the right places, and solutions are found that are compatible with highways standards and the needs of different users' (footnote 50: unless, in specific cases, there are clear, justifiable and compelling reasons why this would be inappropriate)*
- *180 (c) 'development resulting in the loss or deterioration of irreplaceable habitats (such as ancient woodland and ancient or veteran trees) should be refused, unless there are wholly exceptional reasons (footnote 63) and a suitable compensation strategy exists';*

  and provides specific guidance that:

- *180 (d) 'development whose primary objective is to conserve or enhance biodiversity should be supported; while opportunities to incorporate biodiversity improvements in and around developments should be encouraged, especially where this can secure measurable net gains for biodiversity'.*

2.4 With reference to paragraph 180 (c), examples of what is deemed to be *'wholly exceptional'* are included within Footnote 63 and provides the examples of '*infrastructure projects (including nationally significant infrastructure projects, orders under the Transport and Works Act and hybrid bills), where the public benefit would clearly outweigh the loss or deterioration of habitat'*.

Statutory Considerations

2.5 Local authorities have a Duty under the Town and Country Planning Act to create Tree Preservation Orders (TPO) in order to protect and preserve specific trees and woodlands that bring significant amenity benefit to a particular site or location. Under a TPO it is a criminal offence to cut down, top, lop, uproot or wilfully destroy a tree protected by that Order, or to cause or permit such actions, if carried out without the prior written consent of the acting LPA. Anyone found guilty of such an offence is liable and in serious cases, may result in prosecution and incur an unlimited fine.

2.6 Following consultation with the Local Planning Authority, Swindon Borough Council, it is understood that there is a Tree Preservation Order, namely TPO 2 / 2015, which applies to a number of trees present within the wider site but no trees within the Old Vicarage Lane and A240 Junction site boundary are subject to Tree Preservation Orders.

2.7 Information provided on Tree Preservation Orders and Conservation Areas is accurate to the date of this assessment and cannot be assumed to remain unchanged. The last check was carried out on the 26.02.2015.

Non-Statutory Considerations

2.8 In order to compile existing baseline information on relevant arboricultural considerations information was requested from both statutory and non-statutory nature conservation organisations. The Multi Agency Geographic Information for the Countryside (MAGIC)[1] website highlighted no tree cover within the Old Vicarage Lane site as or included within the following:

- The Priority Habitat Inventory, Deciduous Woodland
- The National Forestry Inventory

3.0 SURVEY METHODOLOGY

3.1 The survey of trees has been carried out in accordance with the criteria set out in Chapter 4 of BS5837. The survey has been undertaken by a suitably qualified and experienced arboriculturist and has recorded information relating to all those trees within the site and those adjacent to the site which may be of influence to any proposals. Trees were assessed for their arboricultural quality and benefits within the context of the proposed development in a transparent, understandable and systematic way.

3.2 Due to the time that has lapsed since the original tree survey in 2017, a resurvey of all the trees and hedgerows within the vicinity of the proposed junction was conducted, reviewing their condition and updating previous survey information accordingly. The resurvey of trees was undertaken in January 2022. This has ensured that all the trees and hedgerows within the vicinity of the proposed access which would be affected by the works have been identified.

3.3 Trees have been assessed as groups or hedgerows where it has been determined appropriate.

[1] http://magic.defra.gov.uk/

- The term group has been applied where trees form cohesive arboricultural features either aerodynamically, visually or culturally including biodiversity or habitat potential for example parkland or wood pasture.
- For the purposes of this assessment, a hedgerow is described as any boundary line of trees or shrubs less than 5m wide at the base and are managed under a regular pruning regime.

3.4 An assessment of individual trees within groups or hedgerows has been made where a clear need to differentiate between them, for example, in order to highlight significant variation between attributes including physiological or structural condition or where a potential conflict may arise.

Ancient and Veteran Trees

3.5 Veteran trees and Ancient Woodland are important components of the landscape, their importance can be for a number of reasons including that of their ecological, social, cultural and historic value.

3.6 Veteran Trees and Ancient Woodlands are material considerations within the planning process and their importance is specifically recognised within the National Planning Policy Framework (NPPF) 2021, which defines the terms ancient or veteran tree as:

'A tree which, because of its age, size and condition, is of exceptional biodiversity, cultural or heritage value. All ancient trees are veteran trees. Not all veteran trees are old enough to be ancient, but are old relative to other trees of the same species. Very few trees of any species reach the ancient life-stage.'[2]

3.7 Various published methodologies are currently available which, due to the complexity and subjectivity of the process of defining and assessing these trees, often have conflicting definitions. This assessment, and the criteria used for defining ancient/veteran trees and the identification of attributable ancient/veteran features, has been based on a range of currently published guidance and resources.

BS5837 Categories

3.8 Trees have been divided into one of four categories based on Table 1 of BS5837, *'Cascade chart for tree quality assessment'*. For a tree to qualify under any given category it should fall within the scope of that category's definition (see below).

3.9 Category U trees are those which would be lost in the short term for reasons connected with their physiology or structural condition. They are, for this reason not considered in the planning process on arboricultural grounds. Categories A, B and C are applied to trees that should be of material considerations in the development process. Each category also having one of three further sub-categories (i, ii, iii) which are intended to reflect arboricultural, landscape and cultural or conservation values accordingly.

3.10 **Category (U) – (Red):** Trees which are unsuitable for retention and are in such a condition that they cannot realistically be retained as living trees in the context of the current land use for longer than 10 years. Trees within this category are:

[2] Ministry of Housing, Communities and Local Government. (2019). *National Planning Policy Framework.* London: Ministry of Housing, Communities and Local Government.

- Trees that have a serious irremediable structural defect such that their early loss is expected due to collapse and includes trees that will become unviable after removal of other category U trees.
- Trees that are dead or are showing signs of significant, immediate or irreversible overall decline.
- Trees that are infected with pathogens of significance to the health and/ or safety of other nearby trees or are very low quality trees suppressing adjacent trees of better quality.
- Certain category U trees can have existing or potential conservation value which may make it desirable to preserve.

3.11 **Category (A) – (Green):** Trees that are considered for retention and are of high quality with an estimated remaining life expectancy of at least 40 years with potential to make a lasting contribution. Such trees may comprise:

- Sub category (i) trees that are particularly good examples of their species, especially if rare or unusual, or are essential components of groups such as formal or semi-formal arboricultural features for example the dominant and/or principal trees within an avenue.
- Sub category (ii) trees, groups or woodlands of particular visual importance as arboricultural and / or landscape features.
- Sub category (iii) trees, groups or woodlands of significant conservation, historical, commemorative or other value for example veteran or wood pasture.

3.12 **Category (B) – (Blue):** Trees that are considered for retention and are of moderate quality with an estimated remaining life expectancy of at least 20 years with potential to make a significant contribution. Such trees may comprise:

- Sub category (i) trees that might be included in category A but are downgraded because of impaired condition for example the presence of significant though remediable defects, including unsympathetic past management and storm damage.
- Sub category (ii) trees present in numbers, usually growing as groups or woodlands, such that they attract a higher collective rating than they might as individuals or trees occurring as collectives but situated so as to make little visual contribution to the wider locality.
- Sub category (iii) trees with material conservation or other cultural value.

3.13 **Category (C) – (Grey):** Trees that are considered for retention and are of low quality with an estimated remaining life expectancy of at least 10 years or young trees with a stem diameter below 150mm. Such trees may comprise:

- Sub category (i) unremarkable trees of very limited merit or such impaired condition that they do not qualify in higher categories.
- Sub category (ii) trees present in groups or woodlands, but without this conferring on them significantly greater collective landscape value or trees offering low or only temporary / transient screening benefits.
- Sub category (iii) trees with no material conservation or other cultural value.

Site Plans

3.14 The individual positions of trees and groups have been shown on the Tree Survey Plan. The positions of trees are based on a topographical / land survey, as far as possible, supplied by the client. Where topographical information has not identified the position of trees these have been plotted using a global positioning system and aerial photography to provide approximate locations. The crown spread, root protection area and shade pattern (where appropriate) are also indicated on this plan.

3.15 As part of this assessment, a Tree Retention Plan (1796-T-08) has been prepared to show the proposed layout in relation to the existing tree cover allowing an assessment of any potential conflicts. The plan also identifies which trees would be required to be removed or retained as part of the proposed access.

3.16 A Tree Protection Plan (1796-T-09) has also been included which outlines proposed tree protection measures to be enacted during the construction phase.

Tree Constraints and Root Protection Areas

3.17 Below ground constraints to future development are represented by tree roots and the soil environment in which they grow which needs to be protected if the tree is to be retained. Tree rooting systems are essential for the uptake of water and nutrients, serving the storage of carbohydrates for the future growth and function of the tree, and form structural anchorage and support for the stem and crown. The perceived rooting area of the tree; referred to as the root protection area (RPA) needs to be protected if the tree is to be retained.

3.18 The RPA is a notional area considered to be the minimum zone that must be protected to avoid any adverse impacts on retained trees. The RPA has been calculated in accordance with Annex C, D and Section 4.6 of BS5837:2012 and requires suitable protection in order for the tree to be successfully incorporated into any future scheme. As such, the RPA of existing trees is an important material consideration when considering site constraints and planning development activities.

3.19 Where applicable the shape of the Root Protection Area has been modified to consider the presence of any nearby obstacles (existing or past) which may have restricted root growth and the likely root distribution i.e. the presence of hard standing, structures and underground apparatus. Where groups of trees have been assessed, the Root Protection Area has been shown based on the maximum sized tree in any one group and so may exceed the Root Protection Area required for some of the individual specimens within the group. Further detailed inspection of the individual trees forming a group may be required where development impacts upon the group.

3.20 Whilst it is generally accepted that a trees roots may extend far greater distances than the notional RPA, with the distribution of the root system relating directly to the availability of suitable conditions for growth (namely oxygen, water and nutrients), with roots predominantly located in the upper 1,000 mm of the soil horizon; the RPA offers an accepted protective buffer from development.

3.21 Above ground constraints such as the current crown spread of the trees and an illustration of the shade pattern (where appropriate) have been considered and identified within the Tree Survey Plan and Tree Retention Plan indicates their potential area of shading influence.

Considerations and Limitations of the Tree Survey

3.22 The survey was completed from ground level only and from within the boundary of the site. Aerial tree inspections or an assessment of the internal condition of the stem/s or branches were not undertaken at this stage as this level of survey is beyond the scope of the initial assessment.

3.23 The statements made in this report regarding the assessed trees does not take into account the effects of extreme / adverse weather conditions, changes in land use prior to the site's development, unforeseen accidents or anti-social behaviours, such as vandalism, which occur since the date of the survey. As such, the assessment of tree condition given within applies to the date of survey and cannot be assumed to remain unchanged.

3.24 It will be necessary to review all comments and observations made within this report, in accordance with sound arboricultural practice, within two years of the date of survey (unless explicitly stated elsewhere within this report). Further review may also be necessary where site conditions change or works to trees are carried out which have not been specified in detail within this report.

3.25 Hedgerows are identified as a Habitat of Principal Importance (HPI) as listed within Section 41 of the Natural Environment and Rural Communities (NERC) Act 2006. The tree survey conducted, in accordance with BS5837, does not assess hedgerows against the Hedgerow Regulations 1997 or specifically from an ecological perspective, and is outside the scope of this assessment.

3.26 It may be necessary during detailed design to undertake further assessment and accurate positioning of woody species within tree groups and hedgerows to assist structural calculations for foundation design of structures in accordance with current building regulations. The exact position of individual trees or species included as part of a tree group should be checked and verified on site prior to any decisions for foundation design, tree operations or construction activity being undertaken. Further survey work would be required for calculating foundation depths in accordance with NHBC Chapter 4.2 Building near Trees.

4.0 RESULTS

4.1 A total of eleven individual trees, four group of trees and five hedgerows were surveyed as part of the Arboricultural Assessment. Trees were surveyed as individual trees and groups of trees where examples are clearly present as per the description. Refer to the Tree Survey Plan and Appendix A – Tree Schedule for full details of the trees included in this assessment. The table below summarises the trees assessed.

Tree Schedule

4.2 Appendix A provides a schedule of trees and hedgerows for the wider site but only those trees within the vicinity of the proposed junction have been reviewed and information pertaining to those trees updated as required. All other trees subject to the original survey (except for those

affected by the two remaining points of access and Haul Route, covered by separate Arboricultural Impact Assessment Reports) have not been reviewed at this stage and information pertaining to those trees in the Tree Schedule within the wider site remain as per the 2017 Tree Survey.

4.3 Appendix A presents details of any individual trees, groups and hedgerows found during the assessment including heights, diameters at breast height, crown spread (given as a radial measurement from the stem), age class, comments as to the overall condition at the time of inspection, BS5837 category of quality and suitability for retention and the root protection area.

4.4 General observations particularly of structural and physiological condition for example the presence of any decay and physical defect and preliminary management recommendations have also been recorded where appropriate.

Results Summary

Table 1: Summary of Trees by Retention Category

	Individual Trees	Total	Groups of Trees & Hedgerows	Total
Category U - Unsuitable	T66	1		0
Category A (High Quality / Value)	T65	1		0
Category B (Moderate Quality / Value	T67, T59, T62, T205	4	G32	1
Category C (Low Quality / Value)	T60, T61, T63, T64, T204	5	G31, G33 G34, H28, H29, H30, H32, H88	8

Ancient and Veteran Trees

4.5 None of the assessed trees were considered as ancient or veteran trees in accordance with accepted methodologies and guidance.

5.0 ARBORICULTURAL IMPACT ASSESSMENT

5.1 The following paragraphs present a summary of the tree survey and discussion of particular trees and groups recorded in the context of any proposed development in the form of an Arboricultural Impact Assessment in accordance with section 5.4 of BS5837. Any final tree retentions will need to be reconciled with the advice contained within this report.

5.2 The AIA has been based upon the Drawing Title, '1335-HL-058-107F Planning Layout' produced by Brookbanks and seeks to outline the relationship between the proposals and the existing trees, groups and hedgerows. The drawing shows the proposals for a new junction between Old Vicarage Lane and the A420. An overlay of the layout has been incorporated in the Tree Retention (FPCR 1796-T-08) Tree Protection (FPCR1796-T-09) Plans to assist in identifying the relationship and any potential conflicts between the proposals and the existing trees and hedgerows.

Table 3: Summary of Impact on Tree Stock

	Trees to be Retained as part of the A420 Junction Application	Total	Trees to be Removed in full or part for the A420 Junction Application	Total
Category U (Unsuitable)		0	T66	1
Category A (High Quality / Value)	T65	1		0
Category B (Moderate Quality / Value	T67, T59, T62, T205, G32	5		0
Category C (Low Quality / Value)	T60, T61, G33, G34, H88	4	T63, T64, T204, H28, H29, H30, H32, G31	8

5.3 The A420 Junction proposals will necessitate the removal of the following:

Trees

5.4 Four trees require removal to facilitate the junction proposals, one category U – see below comment - and three category C.

5.5 T66, a category U crack willow *Salix fragilis* was located outside of the proposals and therefore would not need to be removed to directly facilitate the junction upgrade, but due to the category U status, this specimen would be recommended for removal on arboricultural grounds in the interests of safety.

5.6 T63 and T64 were category C ash *Fraxinus excelsior* specimens located within H29, close to the A420. These specimens were old coppice stools which displayed significant seed loading and epicormic growth, likely symptoms of ash dieback and indications of declining condition.

5.7 T204, an English elm located immediately south of a small drainage ditch on the verge to the west of the site.

Hedges

5.8 Four hedgerows will require either full or partial removal to facilitate the junction proposals, all of which were category C due to their size and condition from an arboricultural perspective at the time of survey.

5.9 H28 was a small hedge which ran underneath G31 on the south side of Old Vicarage Lane.

5.10 H29, located to the south of the railway line, this hedge was situated on the northern side of Old Vicarage Lane. It contained a number of elm *Ulmus procera* specimens but at the time of survey had been kept trimmed and so these specimens had not succumbed to Dutch elm disease.

5.11 H30 was located along the north side of the A420, directly north of the drainage ditch. This hedgerow was regularly maintained which gave the hedge a consolidated form.

5.12 H32 was a regularly maintained roadside hedgerow to the north of the railway line, within it contained a number of dead English elm specimens.

Groups

5.13 A single group, G31 will require the southern-most seven metres to be removed to facilitate the proposals. G31, was a category C group containing crack willow, English elm and ash generally in fair condition with some suppressed specimens due to their proximity to one another and dense ivy growth.

Supervised Uplift of Tarmac within the RPA of T61 and T62

5.14 T61 and T62 have a portion of their Root Protection Area located within proposed construction areas. Currently the northeast side of these specimen's RPA is covered with tarmac. As the tarmac in that area will be replaced, but the subbase left in place, it is believed that limited to no disturbance will occur to any roots present in this area. However, as a precaution, it is proposed to provide arboricultural supervision of the works while being carried out.

5.15 The area which requires arboricultural supervision is shown on the Tree Protection Plan (1796-T-09) as a yellow cross-hatch.

Discussion

5.16 In conclusion for arboriculture, the proposals are considered to meet the aims and objectives of national policy through careful consideration of the design and the seeing retention of as much of the existing tree and hedgerow cover as possible whilst allowing for the proposed junction access. The wider development proposals will provide extensive new landscaping including tree and hedgerow planting around all of the new access point thus mitigating the loss of tree and hedgerow cover to the proposals.

Tree Management

5.17 All retained trees should be subjected to sound arboricultural management as recommended within section 8.8.3 of BS5837 *Post Development Management of Existing Trees,* where there is a potential for public access in order to satisfy the landowner's duty of care. Additionally, inspections annually and following major storms should be carried out by an experienced arboriculturist or arborist to identify any potential public safety risks and to agree remedial works as required.

5.18 All tree works undertaken should comply with British Standard 3998:2010 and should therefore be carried out by skilled tree surgeons. It would be recommended that quotations for such work be obtained from Arboricultural Association Approved Contractors as this is the recognised authority for certification of tree work contractors.

5.19 All vegetation and, particularly, woody vegetation proposed for clearance should be removed outside of the bird-breeding season (March - September inclusive) as all birds are protected under the Wildlife and Countryside Act, 1981 (as amended) whilst on the nest. Where this is not possible, vegetation should be checked for the presence of nesting birds prior to removal by an experienced ecologist.

General Design Principles in Relation to Retained Trees

5.20 The routing of below ground services should also be considered with regard to the retained trees. As recommended by the guidance given in section 7.7 of BS5837 services, where possible, should not encroach within the Root Protection Areas of retained trees. If below-ground services are proposed within a Root Protection Area, modifications to the alignment of the service route may need to be made in order to minimise adverse effects on root stability and overall tree health.

5.21 Consideration may also need to be given to the potential for tree roots of newly planted trees and hedgerows to affect or compromise the future services. As far as feasible, it would be preferable that proposed services near both the existing and any new planting should be ducted for ease of access and maintenance and grouped together to minimise any future disturbance.

6.0 NEW TREE AND HEDGEROW PLANTING

6.1 As part of the development proposals new tree planting will be proposed which will more than mitigate the losses required for the Old Vicarage Lane proposals.

6.2 As part of the mitigation for the losses of trees, it would be proposed that replacement planting would be included within public open space within the wider development.

6.3 Species composition and locations for all trees, including those replacement trees for TPO losses, will be determined within future landscaping proposals.

6.4 Where Tree Preservation Order trees are removed, the replacement of these trees will be agreed and approved by the LPA as part of the wider landscaping proposals. No trees with Tree Protection Orders were present within the boundary for the Old Vicarage Lane and A420 Junction works.

Trees

6.5 The landscaping scheme should consider the use of both native tree species (for their low maintenance requirements and nature conservation value) and ornamental species (for their contribution to urban design and amenity value). Species choices should be selected on the basis of their suitability for the final site use. Furthermore, during the design process consultation should be made with the Local Planning Authority to obtain information on their tree strategy and incorporate the planting proposals with any local policies and initiatives and/or Biodiversity Action Plans (BAP).

6.6 In line with the NPPF all schemes should aim achieve a net gain in biodiversity value. Nationally recognised biodiversity metrics allow for the inclusion of, not limited to, newly planted scattered trees, woodlands and hedgerows as a means of compensating for loss of habitat as part of the development. Tree and shrub planting can therefore be used to contribute to this biodiversity gain.

6.7 To maximise biodiversity value (and contribution to net gain) native species or varieties should be specified. Such provisions can be incorporated into both the hard and soft landscaping of the

scheme. It is recommended that tree and hedgerow specifications are made following consultation with guidance published by the Local Planning Authority.

6.8 When deciding upon suitable tree species, careful consideration would need to be given to the following: ultimate height and canopy spread, form, habit, density of crown, potential shading effect, colour, water demand, soil type and maintenance requirements in relation to both the built form of the new development and existing properties.

6.9 Through careful species selection, the landscape scheme shall reduce the risk of trees being removed in the future on the grounds of nuisance. Nuisance can be perceived in a number of ways and vary from person to person however most commonly, within the context of trees, low overhanging branches, excessive shading, seasonal leaf fall and the misinformed perception that trees close to buildings cause damage.

Hedgerows

6.10 Hedgerows are identified as a Habitat of Principle Importance (HPI) as listed within Section 41 of the Natural Environment and Rural Communities (NERC) Act 2006. Consequently, it is important that the proposed scheme delivers a net gain in terms of linear hedgerows through new planting to compensate for any losses. Species should be native, and characteristic of the locality.

6.11 Recommended species for native hedgerow planting are as follows:

- *Crataegus monogyna*
- *Prunus spinosa*
- *Cornus sanguinea*
- *Corylus avellana*
- *Acer campestre*
- *Euonymus europaeus*

General Planting Recommendations

6.12 Wherever possible, following discussions with the developer and utility companies, common service trenches should be specified to minimise land take associated with underground service provision and facilitation access for future maintenance.

6.13 Tree planting should be avoided where they may obstruct overhead power lines or cables. Any underground apparatus should be ducted or otherwise protected at the time of construction to enable trees to be planted without resulting in future conflicts.

7.0 TREE PROTECTION MEASURES

7.1 Retained trees will be protected from construction work by Tree Protection Fencing as shown in the Tree Protection Plan (FPCR 1796-T-09) and in accordance with BS5837 (2012). Appendix B provides the specification of tree protective fencing that will be used.

7.2 Supervision of the uplift of tarmac within the RPA of T61 and T62 will ensure that impacts to retained trees are kept to a minimum and any roots encountered will be appropriately dealt with.

General Information and Recommendations

7.3 All trees retained on site will be protected by suitable barriers or ground protection measures around the calculated RPA, crown spread of the tree or other defined constraints of this assessment as detailed by section 6 and 7 of BS5837.

7.4 Barriers will be erected prior to commencement of any construction work and before demolition including erection of any temporary structures. Once installed, the area protected by fencing or other barriers will be regarded as a construction exclusion zone. Fencing and barriers will not be removed or altered without prior consultation with the Project Arboriculturist.

7.5 Any trees that are not to be retained as part of the proposals should be felled prior to the erection of protective barriers. Particular attention needs to be given by site contractors to minimise damage or disturbance to retained specimens.

7.6 Where it has been agreed, construction access may take place within the root protection area if suitable ground protection measures are in place. This may comprise single scaffold boards over a compressible layer laid onto a geo-textile membrane for pedestrian movements. Vehicular movements over the root protection area will require the calculation of expected loading and the use of proprietary protection systems.

7.7 Confirmation that tree protective fencing or other barriers have been set out correctly should be gained prior to the commencement of site activity.

Tree Protection Barriers

7.8 Tree protection fencing should be fit for the purpose of excluding any type of construction activity and suitable for the degree and proximity of works to retained trees. Barriers must be maintained to ensure that they remain rigid and complete for the duration of construction activities on site.

7.9 In most situations, fencing should comprise typical construction fencing panels attached to scaffold poles driven vertically into the ground. For particular areas where construction activity is anticipated to be of a more intense nature, supporting struts, acting as a brace should be added and fixed into position through the application of metal pins driven into the ground to offer additional resistance against impacts.

7.10 Where site circumstances and the risk to retained trees do not necessitate the default level of protection an alternative will be specified appropriate to the level / nature of anticipated construction activity. The recommended methods of fencing specifications for this site have been illustrated in Appendix B.

7.11 It may be appropriate on some sites to use temporary site offices, hoardings and lower level barrier protection as components of the tree protection barriers. Details of the specific protection barriers for the site can be provided should the application be approved, as part of a site specific Arboricultural Method Statement for a Reserved Matters application and in accordance with the guidance contained within BS5837.

Protection outside the exclusion zone

7.12 Once the areas around trees have been protected by the barriers, any works on the remaining site area may be commenced providing activities do not impinge on protected areas.

7.13 All weather notices should be attached to the protective fencing to indicate that construction activities are not permitted within the fenced area. The area within the protective barriers will then remain a construction exclusion zone throughout the duration of the construction phase of the proposed development. Protection fencing signs can be provided upon request.

7.14 Wide or tall loads etc should not come into contact with retained trees. Banksman should supervise transit of vehicles where they are in close proximity to retained trees.

7.15 Oil, bitumen, cement or other material that is potentially injurious to trees should not be stacked or discharged within 10m of a tree stem. No concrete should be mixed within 10m of a tree. Allowance should be made for the slope of ground to prevent materials running towards the tree.

7.16 No fires will be lit where flames are anticipated to extend to within 5m of tree foliage, branches or trunk, taking into consideration wind direction and size of fire.

7.17 Notice boards, telephone cables or other services should not be attached to any part of a retained tree.

7.18 Any trees which need to be felled adjacent to or are present within a continuous canopy of retained trees, must be removed with due care (it may be necessary to remove such trees in sections).

Protection of Trees Close to the Site

7.19 A number of trees were located on the boundaries of the site and therefore the root protection area and crown spread of these trees will need to be protected in the same way as all the retained trees within the site. All trees located outside the boundaries of the assessment site yet within close proximity to works should be adequately protected during the course of the development by barriers or ground protection around the calculated root protection area.

7.20 Any trees which are to be retained and whose Root Protection Areas may be affected by the development should be monitored, during and after construction, to identify any alterations in quality with time and to assess and undertake any remedial works required as a result.

Protection for Aerial Parts of Retained Trees

7.21 Where it is deemed necessary to operate wide or tall plant within close proximity to trees it is best advised that appropriate, but limited tree surgery, be carried out beforehand to remove any obstructive branches as any such equipment would have potential to cause damage to parts of the crown material, i.e. low branches and limbs, of retained trees within the protective barriers. This is termed as ‘access facilitation pruning’ within BS5837. Any such pruning should be undertaken in accordance with a specification prepared by an arboriculturist.

7.22 A pre-commencement site meeting with contractors who are responsible for operating machinery is advised to firstly highlight the potential for damage occurring to tree crowns and to ensure that extra care is applied when manoeuvring machinery during such operations within close proximity to retained trees to avoid any contact.

7.23 In the event of having caused any branch or limb damage to retained trees it is strongly recommended that suitable tree surgery be carried out, in accordance with British Standard

3998:2010 and in agreement with the Local Planning Authority prior to correcting the damage, upon completion of development.

8.0 CONCLUSION

8.1 The proposed Junction of Old Vicarage Lane and the A420 will require the removal of four individual trees, sections from four hedgerows and a partial removal of a single group to facilitate the proposals.

8.2 Whilst the proposals will require the loss of these trees, groups and hedgerows, they will be more than mitigated for through new high-quality tree and hedgerow planting across the wider development as part of the extensive Green Infrastructure.

8.3 Protection for retained trees surrounding the proposals will be provided in the form of tree protection fencing. An area of tarmac uplift within the RPAs of T61 and T62 will be required, this work will be supervised by the Arboricultural Clerk of Works to reduce the impacts to these trees.

FIGURES

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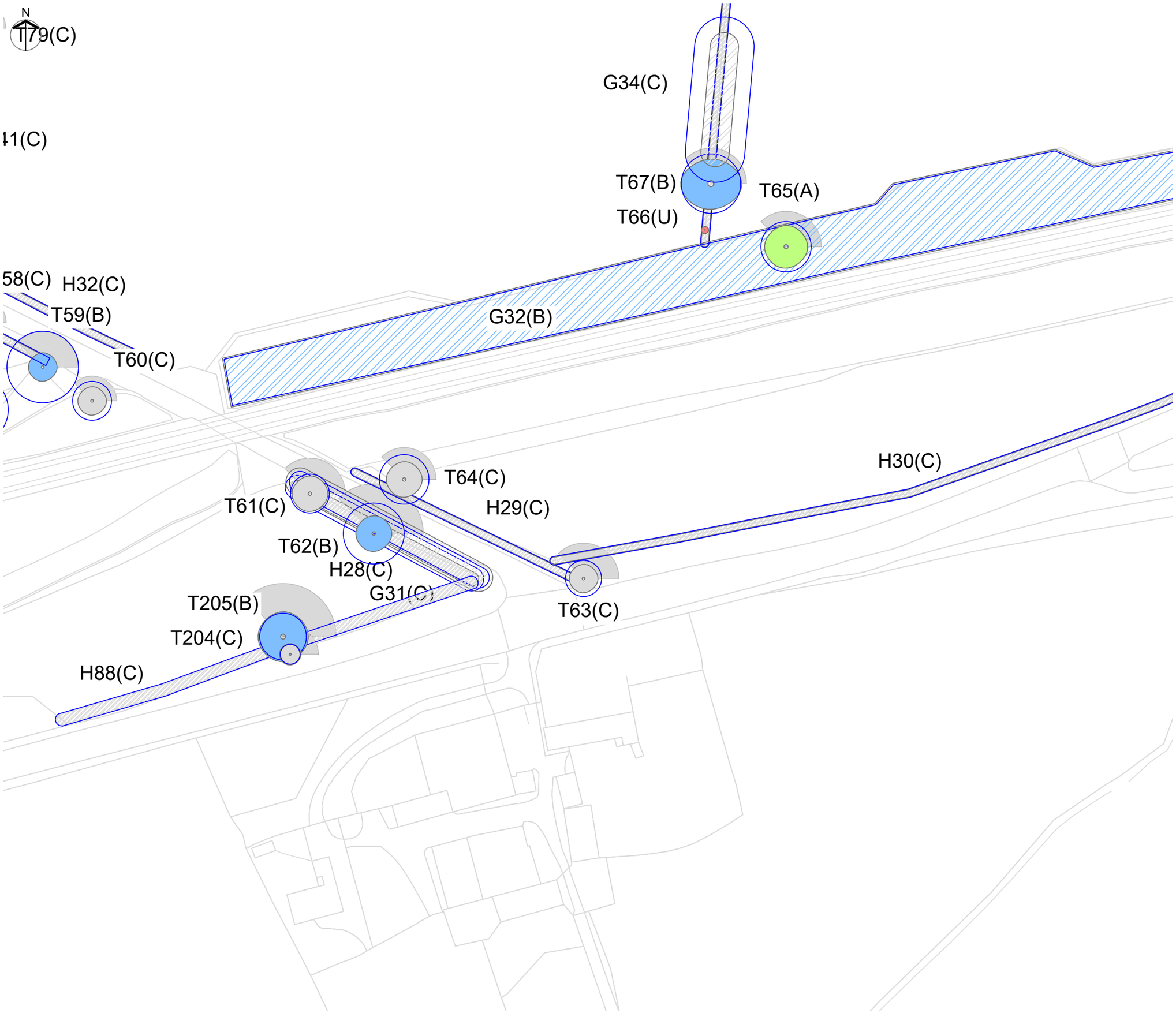

KEY

- Category U - Trees / Groups Unsuitable for Retention (BS 5837:2012)
- Category A - Trees / Groups of High Quality (BS 5837:2012)
- Category B - Trees / Groups of Moderate Quality (BS 5837:2012)
- Category C - Trees / Groups of Low Quality (BS 5837:2012)
- Hedgerow (Colour indicates BS5837:2012 Category)
- Root Protection Area (The RPA has been altered where appropriate to reflect underground constraints)
- T1 (A) G1 (A) — Individual / Group Number and BS5837:2012 Category
- Indicative Shade Pattern (in accordance with BS5837:2012 where appropriate)

Scale 1:1250 @ A3

0 25 50 75m

NOTES

All dimensions to be verified on site. Do not scale this drawing, use figured dimensions only. All discrepancies to be clarified with project Arboriculturalist. Drawing to be read in conjunction with Arboricultural Assessment and Appendix A - Tree Schedule.

Drawing has been produced in colour and is based on digital information in .dwg format, aerial images and/or GPS location where appropriate. A monochrome copy should not be relied upon. The exact position of individual trees or species included as part of a tree group, woodland or hedgerow should be checked and verified on site prior to any decisions for foundation design, tree operations or construction activity being undertaken. Further survey work would be required for calculating foundation depths.

Trees are living organisms that change over time, the condition of all trees illustrated herein, are to be checked by the project Arboriculturalist should works commence 12 months after the date of this survey.

SOME TREES MAY BE SUBJECT TO STATUTORY CONSTRAINTS. IT IS THEREFORE ADVISED THAT NO WORKS SHOULD BE UNDERTAKEN TO ANY TREES ILLUSTRATED HEREIN WITHOUT FIRST OBTAINING THE RELEVANT AUTHORISATION TO DO SO UNLESS AGREED AS PER THE APPROVED PLANS THROUGH PLANNING CONSENT.

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rev	date	description	by
-	25.01.2022	First Issue	AG
A	08.06.2022	H33 Update	MHE

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client
Taylor Wimpey

project
Land North of A420 South Marston and Rowborough, Swindon

drawing title
TREE SURVEY PLAN
Junction of Old Vicarage Lane and A420

scale	drawn/checked	date
1:1250 @ A3	AG/ME	January 2022

drawing number **1796-T-07**

rev **A**

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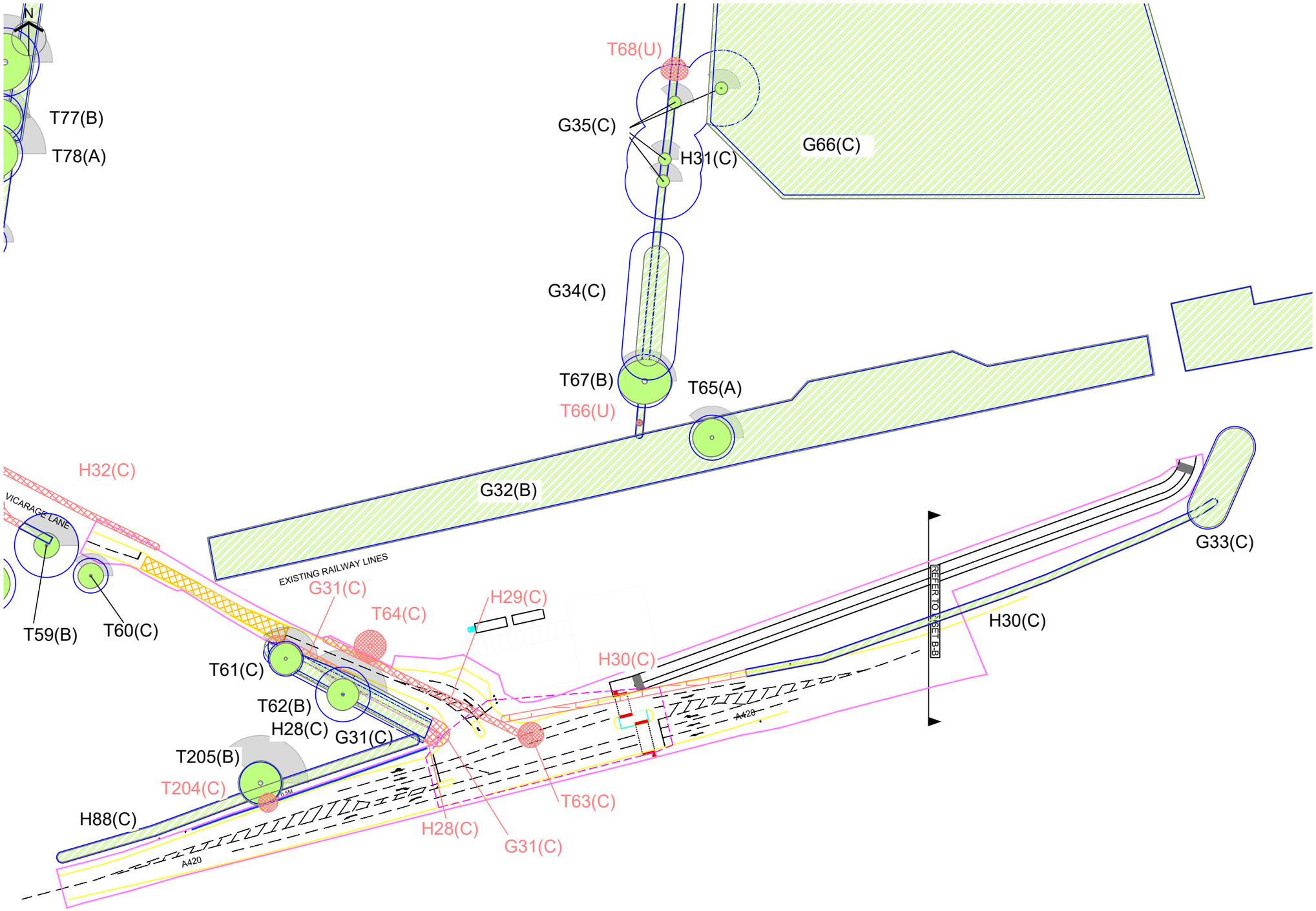

KEY

- Tree/Group to be Retained
- Tree/Group proposed to be removed subject to relevant permissions
- Category U - Unsuitable for retention on arboricultural grounds
- Hedgerow Proposed to be Retained and Incorporated into the New Development
- Hedgerow proposed to be removed subject to relevant permissions
- Root Protection Area (Shown for retained trees only)
- T1 (A) G1 (A) — Individual / Group Number and BS Category
- T1 (A) G1 (A) — Individual / Group Number to be Removed and BS 5837:2012 Category
- Indicative Shade Pattern (in accordance with BS5837:2012 where appropriate)
- A420 Junction Redline Boundary

Scale 1:1250 @ A3

0 25 50 75m

NOTES

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Trees are living organisms that change over time, the condition of all trees illustrated herein, are to be checked by the project Arboriculturalist should works commence 12 months after the date of this survey.

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rev	date	description	by
-	25.01.2022	First Issue	AG
A	03.02.2022	Key Edit	MHE
H	22.08.2022	Layout Update	MHE
I	10.01.2023	Layout Update:1335/HL/58/107F	MHE
J	20.01.2023	Viewport Scale Update	MHE

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client
Taylor Wimpey

project
Land North of A420 South Marston and Rowborough, Swindon

drawing title
TREE RETENTION PLAN
Junction of Old Vicarage Lane and A420

scale	drawn/checked	date
1:1250 @ A3	AG/MHE	January 2022

drawing number
1796-T-08

rev
J

CAD file: S:\1700\1796\ARB\January 2022 - Site Clearance\Plans\Tree Retention Plan - Rev K.dwg

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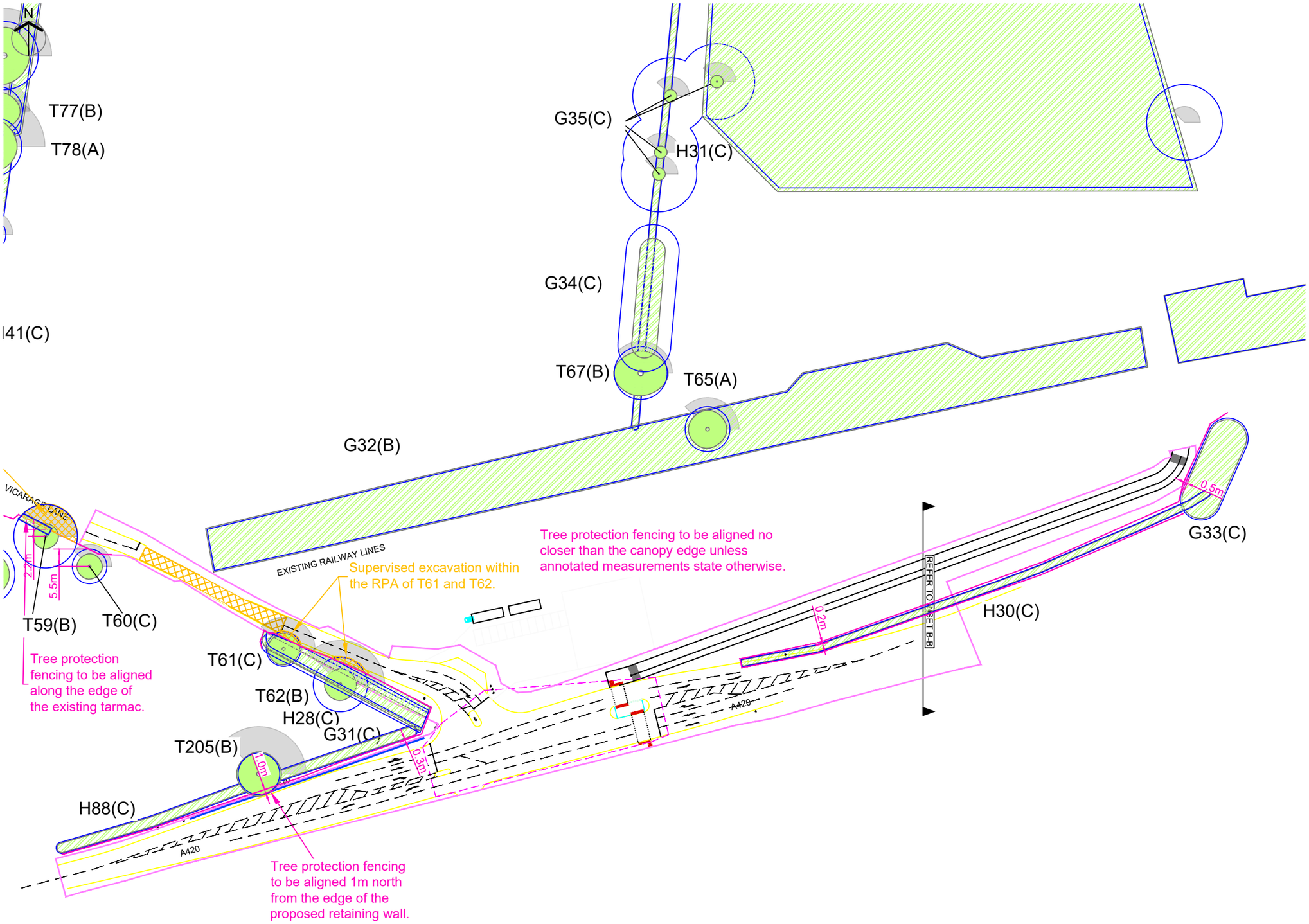

KEY

- Tree/Group to be Retained
- Hedgerow Proposed to be Retained and Incorporated into the New Development
- Root Protection Area (Shown for retained trees only)
- T1 (A) G1 (A) — Individual / Group Number and BS Category
- Indicative Shade Pattern (in accordance with BS5837:2012 where appropriate)
- 6.0m — Line of Protective Barriers (and distance from tree or retained structure)
- Area of Supervised Excavation Within RPA of Retained Trees
- A420 Junction Redline Boundary

Scale 1:1250 @ A3

0 25 50 75m

NOTES

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rev	date	description	by
-	26/01/2022	First Issue	MHE
A	25.02.2022	Layout Update	MHE
H	22.08.2022	Layout Update	MHE
I	10.01.2023	Layout Update:1335/HL/58/107F	MHE
J	20.01.2023	Viewport Scale Update	MHE

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client
Taylor Wimpey

project
Land North of A420 South Marston and Rowborough, Swindon

drawing title
TREE PROTECTION PLAN
Junction of Old Vicarage Lane and A420

scale	drawn/checked	date
1:1250 @ A3	MHE/HCK	January 2022

drawing number **1796-T-09**

rev **J**

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APPENDICES

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Appendix A - Tree Schedule

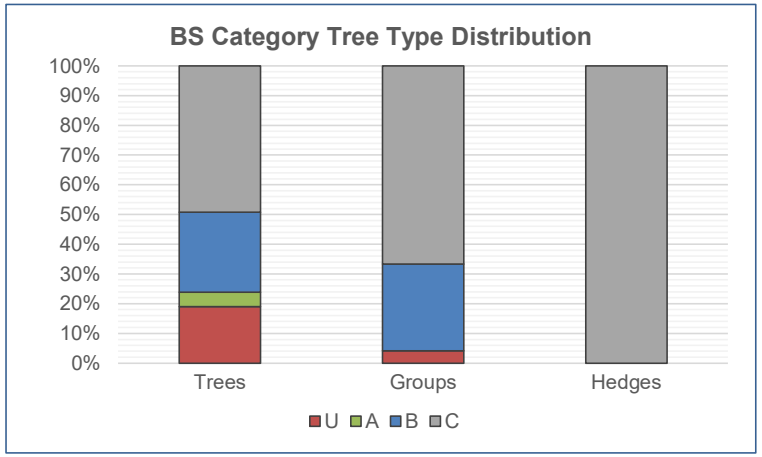
Measurements	Age Classes	Quality Assessment of BS Category	ULE (relates to BS Category)
Height - Measured using a digital laser clinometer (m)	**YNG**: Establishing, typically with good vigour and fast growth rates and strong apical dominance; c. less than 1/3 life expectancy	**Category U -** Trees in such a condition that they cannot realistically be retained as living trees in the context of the current land use for longer than 10 years.	<10 years
Stem Dia. - Diameter measured (mm) in accordance with Annex C of the BS5837	**SM:** Semi-mature trees less than 1/3 life expectancy	**Category A -** Trees of high quality with an estimated remaining life expectancy of at least 40 years.	40+ years
Crown Radius - Measured using a digital laser clinometer radially from the main stem (m)	**EM:** Established, typically vigorous and increasing in apical height and lateral spread; 1/3 - 2/3 life expectancy. Offers landscape significance	**Category B -** Trees of moderate quality with an estimated remaining life expectancy of at least 20 years.	20-40 years
Abbreviations est - Estimated stem diameter avg - Average stem diameter for multiple stems upto - Maximum stem diameter of a group	**M:** Fully established over 2/3 life expectancy, generally good vigour and achieving full height potential with crown still spreading	**Category C -** Trees of low quality with an estimated remaining life expectancy of at least 10 years, or young trees with a stem diameter below 150mm.	10-20 years
	OM: Fully mature, at the extremes of expected life expectancy, vigour decreasing, declining or moribund	Sub-categories: (i) - Mainly arboricultural value (ii) - Mainly landscape value (iii) - Mainly cultural or conservation value	
	V: biological, cultural or aesthetic value comprising niche saproxylic habitat. Individuals of large proportions (stem girth) in comparison to trees of the same species/surviving beyond the typical age range for their species.	**The BS category particular consideration has been given to the following:** • The presence of any structural defects in each tree/group and its future life expectancy • The size and form of each tree/group and its suitability within the context of a proposed development • The location of each tree relative to existing site features e.g. its screening value or landscape features • Age class and life expectancy	

Structural Condition	Physiological Condition
Good - No significant structural defects	**Good -** No significant health problems
Fair - Structural defects that can be remediated	**Fair -** Symptoms of ill-health that can be remediated
Poor - Significant defects beyond remediation, present a risk of failure in the foreseeable future	**Poor -** Significant ill-health. Unlikely the tree will recover in the long term
Dead - Dead tree with structural integrity of tree severely compromised	**Advanced Decline / Dead -** Advanced state of decline and unlikely to recover or Dead

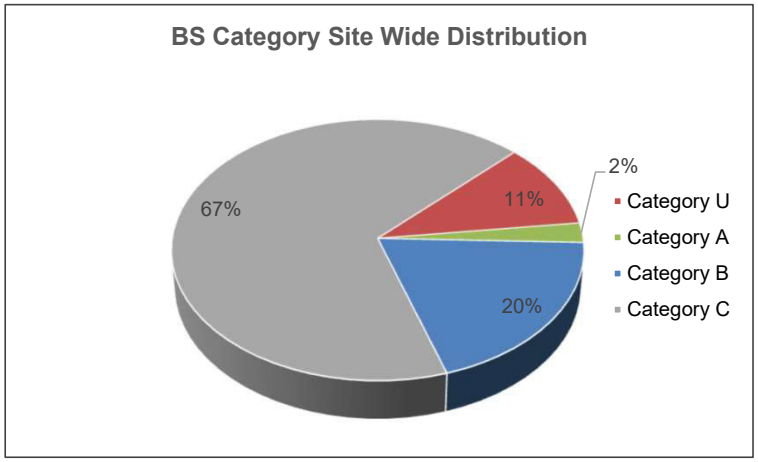
Root Protection Area (RPA)
• The RPA Radius column provides the extent of an equivalent circle from the centre of the stem (m). • The RPA is calculated using the formulae described in paragraph 4.6.1 of British Standard 5837: 2012 and is indicative of the rooting area required for a tree to be successfully retained. Tree roots extend beyond the calculated RPA in many cases and where possible a greater distance should be protected. • Where veteran trees have been identified the RPA has been calculated in accordance with Natural England guidance i.e. 15x the stem diameter, uncapped.

Appendix Summary

	Individual Trees	Totals	Tree Groups and Hedgerows	Totals
Category U	T3, T33, T44, T46, T47, T50, T54, T66, T68, T104, T106, T107	12	G3	1
Category A	T37, T65, T78	3		0
Category B	T21, T34, T38, T53, T55, T56, T57, T59, T62, T67, T73, T74, T75, T76, T77, T102, T205	17	G1, G4, G6, G13, G32, G41, G42	7
Category C	T4, T5, T6, T13, T14, T15, T16, T17, T18, T19, T20, T22, T35, T36, T39, T40, T41, T42, T43, T45, T48, T49, T51, T58, T60, T61, T63, T64, T79, T105, T204	31	G5, G11, G12, G14, G15, G16, G17, G26, G27, G28, G29, G30, G31, G34, G35, G66, H1, H2, H3, H4, H5, H6, H7, H8, H9, H10, H12, H16, H17, H18, H19, H20, H27, H28, H29, H30, H31, H32, H33, H34, H35, H36, H37, H38, H39, H40, H41, H47, H48, H49, H51, H88	52
	Total	63	**Total**	60



BS Category Tree Type Distribution displays the proportion of trees assessed in each type to enable a better understanding of the category distribution.



BS Category Site Wide Distribution shows the proportion of trees assessed in each category across the whole site which allows an interpretation of the site's overall quality.

	Species	Height	Stem Dia.	Crown Radius	Age Class	Overall Condition	Structural Condition	RPA	RPA Radius	BS5837 Cat
INDIVIDUAL TREES										
T3	Crack Willow Salix fragilis	10	500	N - 1.0 S - 7.0 E - 5.0 W - 6.0	M	P	Located within drainage ditch. Multiple stem failure with collapsed stems on ground. Poor form with reduced arboricultural and amenity value. Could be reduced down to hedge level and retained as part of hedge.	N/A	N/A	U
T4	Crack Willow Salix fragilis	12	1300	N - 8.0 S - 9.0 E - 8.0 W - 7.0	OM	P	Old pollarded Willow. Large bole with cavities and cracks which provide bat potential. Part collapsed leaders with well established crown. Recommend to be repollarded back to the old pruning wounds and check for bats prior to tree works.	707	Capped at 15m	C (ii)
T5	Sycamore Acer pseudoplatanus	9	300	3	EM	F	No obvious defects	41	3.6	C (i)
T6	Common Ash Fraxinus excelsior	9	270	N - 4.0 S - 4.0 E - 5.0 W - 4.0	EM	F	Northern ground level side stem	33	3.2	C (i)
T13	Crack Willow Salix fragilis	12	420 310 240 310	8	M	F	Four stems. Pruned on the North side to provide clearance for overhead power lines. Located to East of access drive to farm. Stem bases within 1m of driveway.	193	7.8	C (i)
T14	Crack Willow Salix fragilis	11	420 500	N - 2.0 S - 9.0 E - 7.0 W - 2.0	M	F	Twin stemmed form. Pruned on the South for clearance of overhead power cables. Located to East of access drive to farm. Stem bases within 1m of driveway. Branches pruned for access	193	7.8	C (i)
T15	Crack Willow Salix fragilis	8	230	3	EM	F	Located to East of access drive to farm. Stem bases within 1m of driveway. Branches pruned for access	24	2.8	C (i)
T16	Crack Willow Salix fragilis	12	325 Av	6	M	F	Multiple stems x 7. Located to East of access drive to farm. Stem bases within 1m of driveway. Branches pruned for access	48	3.9	C (i)
T17	Crack Willow Salix fragilis	12	290 360 310 440 240	7	M	F	Multiple stems x 5. Located to East of access drive to farm. Stem bases within 1m of driveway. Branches pruned for access Southern trending leader which bifuricates at 0.5m has failed to the south and it partially hung up in T18. Recommend pollard of whole tree to 2m to reduce the risk of future failure of the remaining leader.	254	9.0	C (i)
T18	Crack Willow Salix fragilis	9	500	3	EM	F	Located to East of access drive to farm. Stem bases within 1m of driveway. Branches pruned for access	113	6.0	C (i)
T19	Crack Willow Salix fragilis	5	330 300 200	N - 4.0 S - 3.0 E - 2.0 W - 1.0	EM	P	Three stems at ground level. Abundance of crown deadwood. Located to East of access drive to farm. Stem bases within 1m of driveway. Branches pruned for access	108	5.9	C (i)

	Species	Height	Stem Dia.	Crown Radius	Age Class	Overall Condition	Structural Condition	RPA	RPA Radius	BS5837 Cat
T20	Common Ash Fraxinus excelsior	8	220 250	N - 3.5 S - 3.0 E - 4.0 W - 3.0	EM	F	Twin stemmed from ground level. Located on bank of drainage ditch High seed loading and epicornmic growth within the crown at the time of resurvey - likely caused by ash dieback.	50	4.0	C (i)
T21	Beech Fagus sylvatica	10	450	6	M	G/F	Good crown/ branch structure with raised crown. Located offsite (private garden) within managed hedgerow	92	5.4	B (i)
T22	Weeping Willow Salix x chrycosoma	14	Est 500	6	M	G	Typical form of species. Located 2m off-site within private garden	113	6.0	C (i)
T33	Common Ash Fraxinus excelsior	7	320 120 120	N - 3.0 S - 4.0 E - 3.5 W - 3.0	M	F	Specimen in extensive decline at time of 2022 resurvey. Re-growth from old coppiced stool following past hedgerow management. Multi stemmed form with three main stems from ground level and forming part of the hedgerow. Small stature but original stool large (approximately 650mm)	N/A	N/A	U
T34	English Oak Quercus robur	15	900	N - 7.0 S - 6.0 E - 6.0 W - 7.0	M	F	Visible deadwood and branch damage within crown. Evidence of possible water-logging ground contibuting to the dieback and amount of accumulated deadwood. Recommendation would be to remove deadwood in the interests of safety	366	10.8	B (i)
T35	English Oak Quercus robur	12	820	N - 7.0 S - 8.0 E - 7.0 W - 7.5	M	F	Tearout wound to southern branch at 4m Crown showing decline Visible deadwood and branch damage within crown. Evidence of possible water-logging ground contibuting to the dieback and amount of accumulated deadwood. Recommendation would be to remove deadwood in the interests of safety	304	9.8	C (i)
T36	Common Ash Fraxinus excelsior	8	220 200	N - 4.0 S - 4.0 E - 2.5 W - 3.0	SM	F	Selfset within hedgeline.Twin stem 1m above ground level with moderate stature. Part suppressed to East by T37. Minor branch damage. Minor deadwood.	40	3.6	C (i)
T37	English Oak Quercus robur	15	955	N - 9.0 S - 7.0 E - 9.0 W - 7.5	M	G	Good crown/branch structure. Minor deadwood (aesthetic only).	413	11.5	A (i)
T38	English Oak Quercus robur	12	950	N - 4.0 S - 5.5 E - 5.0 W - 6.0	M	F	Good crown/branch structure. Minor deadwood and branch damage associated with tree's maturity. Large bole. Established hedgerow tree. Light Ivy cladding on stem. Burring on lower stem	408	11.4	B (i)
T39	Common Ash Fraxinus excelsior	10	Est750	N - 6.0 S - 4.5 E - 5.0 W - 5.5	M	F/P	Main stem located to West of drainage ditch. Crown indicating signs of decline with decay in old branch wounds. Ivy clad with weighted branch over adjacent highway. Storm damage. Fungal bracket (Inonotus Hispidus) on ground. Recommend tree to be pollarded for safety	254	9.0	C (i)

	Species	Height	Stem Dia.	Crown Radius	Age Class	Overall Condition	Structural Condition	RPA	RPA Radius	BS5837 Cat
T40	Common Ash Fraxinus excelsior	9	220 220	N - 5.0 S - 3.5 E - 4.5 W - 3.0	SM	G	Specimen is part of hedgerow and has a form comprising co-dominant stems	44	3.7	C (i)
T41	Common Ash Fraxinus excelsior	8	550	N - 4.0 S - 5.5 E - 4.0 W - 4.0	SM	F	Form comprises a heavily weighted branch to South West. Located within hedgerow. Minor branch degradation. Dense Ivy to 6m	137	6.6	C (ii)
T42	English Oak Quercus robur	14	1200	N - 8.0 S - 6.0 E - 6.0 W - 8.0	M	F	Heavily waterlogged on the northern side at the time of the resurvey. Symmetrical crown structure with visible dieback and major deadwood distribution throughout which may be due to localised waterlogging of the ground. Recommend deadwood to be removed. May need to heavily	651	14.4	C (i)
T43	Common Ash Fraxinus excelsior	6	350	N - 3.0 S - 2.5 E - 3.0 W - 3.0	SM	P	Poor form and small structure with central leader exhibiting signs of decline. Broken branches lower crown	55	4.2	C (ii)
T44	Common Ash Fraxinus excelsior	8	770	N - 5.0 S - 2.0 E - 5.0 W - 3.0	M	P	Old pollarded specimen with signs of general decline and decaying thoughtout with a hollow base. Recommend for removal in the interests of safety (East side overhangs highway)	N/A	N/A	U
T45	Common Ash Fraxinus excelsior	9	680	N - 5.5 S -7.0 E - 5.0 W -6.0	M	P	Decaying leader orientated towards highway. Branch damage with visible fungal bracket (Inonotus hispidus). Stem appears to be hollowing. Recommend repollard to remove the decaying leader on safety grounds	209	8.2	C (ii)
T46	Crack Willow Salix fragilis	9	550 320	N - 4.0 S - 4.0 E - 2.0 W - 6.0	M	P	Main leader failed into the field to the northwest of the stem location. Remove specimen.	N/A	N/A	U
T47	Crack Willow Salix fragilis	2	1000	up to 2m	M	P	Whole stem failed into the field to the northwest Stump with minor regrowth. Recommended for removal	N/A	N/A	U
T48	Crack Willow Salix fragilis	8	220	N - 3.0 S -3.0 E - 3.0 W - 6.0	EM	F	Leaning at 25 degrees to the northwest, recommend coppice to reduce chance of failure. Self set with stem leaning to West. Typical form of species otherwise	22	2.6	C (i)
T49	Crack Willow Salix fragilis	10	Est320	Up to 8	M	P	Collapsed in the past and regrown. Multi stemmed form with three main stems and one cut back on the western side	46	3.8	C (i)
T50	Crack Willow Salix fragilis	13	880	N - 2.0 S - 5.0 E - 7.0 W - 4.0	M	P	Poor form with major failure of main stem. Open wound with structural decay, stem lying on ground. Hollow stem potentially. Recommend pollard to 2m height and remove fallen stem.	N/A	N/A	U
T51	Crack Willow Salix fragilis	11	350 290	N - 5.0 S - 5.0 E - 5.0 W - 6.0	SM	F	Leader snapped 2m above union with stem in field to the northwest, remove failed limb	93	5.5	C (ii)

	Species	Height	Stem Dia.	Crown Radius	Age Class	Overall Condition	Structural Condition	RPA	RPA Radius	BS5837 Cat
T53	English Oak Quercus robur	11	830	6	M	G	Symmetrical crown with good crown/branch structure. Minor deadwood (aesthetic only)	312	10.0	B (i)
T54	English Oak Quercus robur	12+	900	N - 5.0 S - 8.0 E - 6.0 W - 5.0	M	P	Declining form with visible dieback of the crown especially to the North and West sides. Decaying base of the stem with cavities and in particular major root buttress damage on the South West side. Dieback of the crown would be considered as major due to the majority of the upper crown being dead and of stag headed form especially where the northern side is almost entirely dead. Evidence of active crown retrenchment. Also major bark delamination to South and West sides of the stem for almost full height of the stem	N/A	N/A	U
T55	English Oak Quercus robur	14	970	N - 6.0 S - 6.0 E - 7.0 W - 4.0	M	F	Declining form with visible dieback of the crown especially on the upper eastern side which has a stag headed form. Major deadwood and storm damaged branch material. Canopy overhangs highway. Slightly leaning stem to East. Recommend removal of deadwood on safety grounds	426	11.6	B (ii)
T56	English Oak Quercus robur	16	980	N - 5.0 S - 7.0 E - 8.0 W - 7.0	M	G	Symmetrical crown and good balance. Major deadwood on the Northside, with other storm damaged branch material. Ivy clad. Recommend removal of deadwood on safety grounds	434	11.8	B (i)
T57	English Oak Quercus robur	14	950	N - 7.0 S - 7.0 E - 6.0 W - 7.0	M	F	Branch tearout wound at 3m on southern aspect, partially occluded Major deadwood on the Northside and some storm damaged branch material. Root plate compaction from livestock and damage to root buttresses on the South West side. Recommend removal of deadwood on safety grounds	408	11.4	B (i)
T58	English Oak Quercus robur	7	270	4	M	F	Ivy to 6m. Generally suppressed form. Broken branches to road side.	33	3.2	C (i)
T59	English Oak Quercus robur	10	840	4	M	P	Broken branches and pruning wounds to road side. Major deadwood. Eastern crown pruned off. Tree showing signs of distress. Dense Ivy throughout tree. Recommend removal of deadwood on safety grounds	319	10.1	B (i)
T60	Goat Willow Salix caprea	7	460	4	M	F	Typical of species. Off site behind fencing.	96	5.5	C (i)
T61	Common Ash Fraxinus excelsior	10	400 170 146	5	M	F	Old coppiced stool. Multi stemmed form with three stems from ground level. Ivy clad.Some broken branches and root suckers. Dense Ivy throughout specimen. Has a snapped sidestem	95	5.5	C (i)
T62	English Oak Quercus robur	14	720	5	M	F	Densely ivy clad. Minor deadwood throughout.	235	8.6	B (i)

	Species	Height	Stem Dia.	Crown Radius	Age Class	Overall Condition	Structural Condition	RPA	RPA Radius	BS5837 Cat
T63	Common Ash Fraxinus excelsior	10	240 210 190 150 140	4	M	F	Close to A420. Old coppiced stool. Multi stemmed form with five stems from ground level. Has an understory of elm Showing ash dieback symptoms, high seed loading and epicormic growth within the canopy.	81	5.1	C (i)
T64	Common Ash Fraxinus excelsior	9	260 210 480	5	EM	F	Southern leader dead, major deadwood within the canopy overhanging road. Unmanaged hedgerow stands with poor form and group suppression. Set within hedgeline. Possible large old coppice with three main stems on opposite side of ditch. Light ivy. Part of hedgerow H29	155	7.0	C (i)
T65	Hornbeam Carpinus betulus	10	Est 590	6	M	G	On railway embankment. Evenly balanced form and no obvious defects	157	7.1	A (i)
T66	Crack Willow Salix fragilis	10		1	M	P	Collapsed stems and regrowth in Western side (10m).Recommend pollarding in the interests of safety			U
T67	English Oak Quercus robur	10	700	N - 7.0 S - 7.0 E - 8.5 W - 8.5	M	G	Symmetrical crown. Minor deadwood (aesthetic only). Low crown and spreading form. Would recommend crown lifting if needed to suit scheme proposals	222	8.4	B (i)
T68	Common Ash Fraxinus excelsior	12	770	N - 5.0 S - 2.0 E - 4.0 W - 4.5	M	P	Specimen in structural decline with evidence of significant heartwood decay in main stem resulting from failure of the main stem. Dieback and deadwood throughout crown structure. A hollow stem which has an opening leading to internal cavity in a south facing direction. Recommendation would be to pollard at a height of approximately 3m and retain for habitat	N/A	N/A	U
T73	English Oak Quercus robur	11	650	N - 4.0 S - 5.0 E - 7.0 W - 6.0	M	F	Suppressed to South. Minor deadwood. Located on site boundary. Interlocking crown with T74	191	7.8	B (ii)
T74	English Oak Quercus robur	9	430	N - 3.0 S - 7.0 E - 6.0 W - 5.0	M	F	Located on site boundary to West of ditch. Some minor deadwood. Interlocking growth of crown with T73 &74	84	5.2	B (ii)
T75	English Oak Quercus robur	9	400	5	M	F	Form defined by group suppression. Minor deadwood and damaged branches evident. Located at site boundary to West of ditch. Interlocking crown growth with T74	72	4.8	B (ii)
T76	English Oak Quercus robur	15	Est 900	N - 9.0 S - 9.0 E - 8.0 W - 8.0	M	G	Symmetrical crown. Minor deadwood. Located off site west of ditch. Branches overhang into site, prominent specimen	366	10.8	B (i)

	Species	Height	Stem Dia.	Crown Radius	Age Class	Overall Condition	Structural Condition	RPA	RPA Radius	BS5837 Cat
T77	English Oak Quercus robur	10	Est 650	N - 6.0 S - 6.0 E - 7.0 W - 6.0	M	F	Located to West of drainage ditch. Dieback and major deadwood visible. Branches overhang site to East. Recommend removing deadwood overhanging the site on safety grounds	191	7.8	B (ii)
T78	English Oak Quercus robur	17	Est 800	N - 9.0 S - 10.0 E - 9.0 W - 10.0	M	G	Symmetrical crown. Minor deadwood. Located off site. Branches overhang into site up to 4m	290	9.6	A (i)
T79	Common Ash Fraxinus excelsior	8	480	N - 3.5 S - 5.0 E - 4.5 W - 3.0	M	F	Located to West of drainage ditch. Past branch reduction westside Ivy on main stem, specimen showing decline.	104	5.8	C (i)
T102	Horse Chestnut Aesculus hippocastanum	11	650	7	M	G	End tree in a line of trees on private land (Off-site). Three leaders from 2m with a tight union (possible included bark)	191	7.8	B (i)
T104	Common Ash Fraxinus excelsior	2	500	1	M	P	Only stump remaining	N/A	N/A	U
T105	Common Ash Fraxinus excelsior	5	180	2	Yng	F	In hedgerow H49. Typical form of species with no obvious defects. Minor broken branches	15	2.2	C (i)
T106	Common Ash Fraxinus excelsior	3		1	M	P	Failed ivy clad stump. Limited live growth- one branch. Fungal pathagen (Daldinia concentrica) present			U
T107	Common Ash Fraxinus excelsior	7	350	N - 4.5 S - 3.0 E - 4.0 W - 2.0	OM	P	Decayed base which would raise concerns for structural stability. Base of the tree contains a major cavity development at ground level North facing entrance. Small stature and declining form with limited useful life expectancy. Service lines to South. Recommend removal on grounds of safety	N/A	N/A	U
T204	English Elm Ulmus procera	8	240	3	EM	F	Major Pruning wound at 1.5m on southern aspect, partially occluded. Located immediately south of drainage ditch c.4m north of road kerb	26	2.9	C (i)
T205	Goat Willow Salix caprea	15	300 320 250 220	7	M	F	Located 3m north of drainage ditch to the north of the A420 Multileadered from 1m First southern trending branch at 5m	137	6.6	B (i)

	Species	Height	Stem Dia.	Crown Radius	Age Class	Overall Condition	Structural Condition	RPA	RPA Radius	BS5837 Cat
GROUPS OF TREES										
G1	Yew Taxus baccata	8	480	N - 4.0 S - 3.0 E - 2.0 W - 2.0	M	F	Three main trees with one dead tree. Located within garden with fair form. Group suppression to East and West	104	5.8	B (ii)
G3	1 x Apple Malus domestica 2 x Pear Pyrus calleryana Chanticleer	7	400	3	M	P	Only the apple remains of the group, the two pears have snapped main leaders at c.2m One specimen located in the field parcel. Declining condition with decay and structural damage. Apple has fungal pathagen (Inonotus hispidus) and is catorgorised as U	N/A	N/A	U
G4	Silver Birch Betula pendula Weeping Willow Salix x chrycosoma	11	500	Av 3.5	Y-M	F	Ornamental planting located in neighbouring garden. Establishing trees Evidence of management with minimal crown encroachment into site (2m).	113	6.0	B (ii)
G5	Cypress- Chamaecyparis sp.	10	350	3	SM-M	F	Conifer screening located off site in neighbouring garden. Positioned on raised bank. Good form with minor canopy encroachment. 6 trees to the south of TG6 and 2 trees to the north	55	4.2	C (i)
G6	Crack Willow Salix fragilis	12	400 540 400 690	N - 5.5 S - 5.5 E - 9.0 W - 9.0	M	F	Located to West of drainage ditch in part raised position. Evidence of past leader damage with broken branches within crown. Fair condition despite damage. Canopies overhang site to East.One South end twin stem at ground level. Up to 4m crown spread onto site	492	12.5	B (ii)
G11	Cypress- Chamaecyparis sp.	6	250	N - 2.0 S - 2.0 E - 2.5 W - 2.5	SM	G	Conifer screening with encroaching canopies from minimal management	28	3.0	C (i)
G12	Crack Willow Salix fragilis out of site	13	600	5	M	F	Located within garden with one of the group having a poor crown formation due to storm damage. One Willow has been pruned to clear utility lines. 2m off site	163	7.2	C (ii)
G13	1 x Common Ash Fraxinus excelsior 3 x English Oak Quercus robur	10	400	5	M	G	Good condition with minor suppression to group. Located within garden with part managed canopy to south for clearance over adjacent track. Light ivy growth on some stems	72	4.8	B (i)
G14	Cypress- Chamaecyparis sp.	4	200	1.5	M	G	Conifer screening with reduced height to control upward growth. Maintained form	18	2.4	C (i)

	Species	Height	Stem Dia.	Crown Radius	Age Class	Overall Condition	Structural Condition	RPA	RPA Radius	BS5837 Cat
G15	Common Ash Fraxinus excelsior 2 x Horse Chestnut Aesculus hippocastanum Field Maple Acer campestre	up to 8	310	Up to 6	M	F	1 x Large Horse Chestnut - multi stem 250x240x220x200x250 1 x Small Horse Chestnut has failed (U) Ash and Field Maple have typical form of their species with no obvious defects	43	3.7	C (i)
G16	2x Crack Willow Salix fragilis	12	360 310 330 220	Up to 7	OM	P	On opposite side of ditch. Most northerly tree has collapsed at bole with one stem remaining. The group have some storm damage and broken branches and some stems are ivy clad. It would be recommended to pollard the specimens and retain for habitat	173	7.4	C (i)
G17	Crack Willow Salix fragilis approx 5 main trees	12	Av 500	Av 7	M	P	Eastern most individual now removed Second eastern most individual pollarded to 2m with c. 2 year old regrowth present. Unmanaged group of Willow located within/close to drainage ditch. Overall poor with many collapsed leaders and consiquent re-growth. Ivy on many stems. Recommend pollarding as required to reduce risk of future stem failure, especially trees located close to public footpath	113	6.0	C (ii)
G26	2 x Common Ash Fraxinus excelsior 4 x Hornbeam Carpinus betulus	6	200	2-3	EM	F	Along fenceline. Pruned for clearance of branches over the track	18	2.4	C (i)
G27	Crack Willow Salix fragilis Common Ash Fraxinus excelsior	11	Av 500	Up to 6	M	F	Willow group with 1x Ash. Growing within or around pond. Unmanaged form with part collapsed stems. Some of the trees within the group are of poor quality. Providing wildlife habitat	113	6.0	C (ii)
G28	8 x Crack Willow Salix fragilis	Av 17	1000	8	M	F/P	Most of the trees are old pollards. Selected trees have collapsed stems and associated cracks/splits. Some recent branch removals and possible storm damage remediation. Some younger trees have failed also Recommend removing failed stems and undertaking remedial work if public access is to be considered in close proximity	452	12.0	C (ii)
G29	3 x Crack Willow Salix fragilis	Upto 5	Est 1000	7	M	F	Old pollards. Recently collapsed leaving rotting stumps. It would be recommended to pollard the specimens and retain for habitat	452	12.0	C (i)
G30	3 x Hybrid Black Poplar Populus x canadensis	20	760 320	Max 8	M	F	Located at base of railway embankment. Established canopies with minor leader damage and branch shedding. One single stem and two multi-stemmed trees. Some minor storm damage and deadwood	308	9.9	C (i)

	Species	Height	Stem Dia.	Crown Radius	Age Class	Overall Condition	Structural Condition	RPA	RPA Radius	BS5837 Cat
G31	Crack Willow Salix fragilis English Elm Ulmus procera Common Ash Fraxinus excelsior	8	upto 250	Up to 4	M	F	Small copse close to the A420. Generally willow with some elm and one ash which form an understory to several larger trees. Ivy on most stems to 5-6m	28	3.0	C (i)
G32	English Oak Quercus robur Hawthorn Crataegus monogyna (self set) Common Ash Fraxinus excelsior Hornbeam Carpinus betulus	8	Est 300	4	EM	G	Group comprises the material present along the northern slopes of the railway embankment. The hawthorn are established bush forms. Occasional dead trees	41	3.6	B (i)
G34	Common Ash Fraxinus excelsior	9	250 230 180 160	N - 3.0 S - 4.0 E - 4.0 W - 3.5	M	F	Multi stemmed form with evidence of past coppicing. There has been regrowth within the old hedgeline	78	5.0	C (ii)
G35	Crack Willow Salix fragilis	6	Plus 1000	2	OM	F	All specimens have been pollarded in the recent past. Main stems have new re-growth from pollarded cuts. Pollard at below 3m and appears to be regularly undertaken	452	12.0	C (ii)
G41	11 x Common Ash Fraxinus excelsior	plus 14	500	6	M	F	Hedgerow grouping. Overall fair structure and condition with good height clearance site side. Located off site to South of drainage ditch. Some lightly ivy clad stems. Overall, typical form for the species. Some multiple stems but most of single stemmed form	113	6.0	B (i)
G42	6 x Common Ash Fraxinus excelsior 1 x Crab Apple Malus sylvestris 2 x Hawthorn Crataegus monogyna	max 9	Est 500	Up to 5	M	G	Residential gardens. Mixed species with all stems located outside the site boundary by approximately 1-2m. Overhanging as much as 4m in places. One Ash is a large coppice stool. Several Ash in a row, possibly part of an old hedgerow boundary. One large pear (T103). Outgrown hawthorn hedgerow at South end up to 5m. Contains ivy clad stems	113	6.0	B (i)
G66	Crack Willow Salix fragilis	7	Below 150	Up to 3	Yng/EM	F	Large area of willow coppice currently some uncut and some cut Planted as a crop for biomass	10	1.8	C (i)

	Species	Height	Stem Dia.	Crown Radius	Age Class	Overall Condition	Structural Condition	RPA	RPA Radius	BS5837 Cat
HEDGEROWS										
H1	Hawthorn Crataegus monogyna Blackthorn Prunus spinosa Common Ash Fraxinus excelsior	2	est 100	1	M	G	Not maintained north of northern most part of TG3. South of this, regularly trimmed to maintain a consolidated form and shape Roadside position directly adjacent to Thornhill Road Unbroken Some elm present also	5	1.2	C (ii)
H2	Field Maple Acer campestre Hawthorn Crataegus monogyna Elm Ulmus spp. Ash Fraxinus excelsior	1.5	est 100	1	M	G	Historically managed to maintain form and shape Recent flailing evidence on the south side Some elder creeping in Some growth, where less regularly managed has attained up to 4m height - particularly elm and hawthorn	5	1.2	C (ii)
H3	Hawthorn Crataegus monogyna Elder Sambucus nigra Elm Ulmus spp.	5 (max. 7)	est 150	4	M	G	Prominent field boundary hedgerow separating field compartments growing along the line of a ditch course Outgrown form where any attempts at previous management has lapsed over some time - generally no signs of any form of management Occasional self set sycamore and ash standards with separate stems attaining up to 7m height Noted several dead elm	10	1.8	C (ii)
H4	Hawthorn Crataegus monogyna Elm Ulmus spp.	up to 5	est 220	4	M	F	Field boundary Contains a high amount of dead hedgerow elm Outgrown form and gappy form, defunct in places High content of invasive bramble No active signs of management or regular maintenance	22	2.6	C (ii)
H5	Hawthorn Crataegus monogyna Elm Ulmus spp.	1.5	est 100	2	M	F	Short section around property boundary Contains some outgrown elder and elm up to 4m Holm oak specimen at the eastern end at approximately 6m height	5	1.2	C (ii)
H6	Hawthorn Crataegus monogyna	3	est 100	3	M	F	Another short section around property boundary Not maintained and is outgrown Post and rail fence forms the boundary	5	1.2	C (ii)
H7	Bramble Rubus fruiticosa	less than 1	est 100	less than 2	M	P	Position adjacent to the railway line boundary and along the fence line Mostly bramble but contains occasional free standing hawthorn up to approximately 4m height	5	1.2	C (ii)

	Species	Height	Stem Dia.	Crown Radius	Age Class	Overall Condition	Structural Condition	RPA	RPA Radius	BS5837 Cat
H8	Blackthorn Prunus spinosa Hawthorn Crataegus monogyna Elm Ulmus spp	5	est 200	up to 6	M	F	Forms field boundary and is predominantly hawthorn Gappy and outgrown form with no evidence of formal management Post and rail boundary marks the edge of the property Contains numerous dead elm and some elder Ivy present throughout	18	2.4	C (ii)
H9	Hawthorn Crataegus monogyna Elm Ulmus spp Crab Apple Malus sylvestris	6	est 200	up to 6	M	G	Trackside position Evidence of past management to reduce growth over the track through flailing High content of invasive bramble Small quantity of more established crab apple and an occasional field maple	18	2.4	C (ii)
H10	Hawthorn Crataegus monogyna Elm Ulmus spp Crab Apple Malus sylvestris	6	est 200	up to 6	M	G	Substantial boundary feature around field margin containing for the most part an outgrown hedgerow High content of bramble and many dead hedgerow elms Occasional more established crab apple and self set ash elder present too	18	2.4	C (ii)
H12	Hazel Corylus avellana Hawthorn Crataegus monogyna	5	est 150	4	M	F	Property boundary Trees present within the property	10	1.8	C (ii)
H16	Hawthorn Crataegus monogyna Crab Apple Malus sylvestris	5	est 150	4	M	F	Around the property boundary and on the fence line Contains many multiple stemmed specimens resulting from previous management Some elm	10	1.8	C (ii)
H17	Hawthorn Crataegus monogyna Elm/ Crab Apple Malus sylvestris Some substantial Field Maple Acer campestre	5-7	est 200	up to 6	M	F	Field boundary hedgerow and along the course of a ditch Ivy growth in abundance Gappy and outgrown form Contains occasional poplar and many dead hedgerow elm	18	2.4	C (ii)
H18	Hawthorn Crataegus monogyna Crab Apple Malus sylvestris Elm Ulmus spp Small amount of Crack Willow	4-6	est 200	4-6	M	F	Field boundary Defunct form at the northern end High content of ivy throughout Well established structure for most part but of outgrown form Occasional blackthorn noted Some field access gaps	18	2.4	C (ii)

	Species	Height	Stem Dia.	Crown Radius	Age Class	Overall Condition	Structural Condition	RPA	RPA Radius	BS5837 Cat
H19	Elm Ulmus spp Crack Willow Hawthorn Crataegus monogyna	6	est250	8	M	F	Field boundary Gappy and defunct in places, with outgrown form Ivy and dead bramble noted In places, some hawthorn have become more developed to form standards	28	3.0	C (ii)
H20	Hawthorn Crataegus monogyna Field Maple Acer campestre	4-5	est 150	4	M	F	Short section along property boundary Contains occasional ash specimen of more developed form up to 7m	10	1.8	C (ii)
H27	Hawthorn Crataegus monogyna Blackthorn Prunus spinosa Elm Ulmus spp	5	est 150	4	M	F	Field boundary hedgerow A large number of dead elm specimens present No evidence of past management and has become defunct in places	10	1.8	C (ii)
H28	Common Ash Fraxinus excelsior Hawthorn Crataegus monogyna Willow Salix fragilis	8	est 150	4	M	F	Roadside hedgerow of outgrown form No structure and evidence of formal management Contains individual trees Hedgerow for the most part contains hawthorn and forms an under story to the trees above Occasional blackthorn	10	1.8	C (ii)
H29	Hawthorn Crataegus monogyna and small content of Elm Ulmus spp	2	est 100	Less than 2	M	F	Roadside hedgerow Regularly trimmed to maintain a consolidated form and shape Some blackthorn	5	1.2	C (ii)
H30	Hawthorn Crataegus monogyna Elm Ulmus spp	2	est 100	1	M	F	Roadside hedgerow, located immediately north of drainage ditch Regularly trimmed to maintain a consolidated form and shape Some blackthorn The north side currently was unflailed Not as tightly clipped as H29	5	1.2	C (ii)
H31	Hawthorn Crataegus monogyna Blackthorn Prunus spinosa	2	est 100	2	M	G	Internal field boundary hedgerow Regularly trimmed to maintain a consolidated form and shape in places Some blackthorn	5	1.2	C (ii)
H32	Hawthorn Crataegus monogyna Blackthorn Prunus spinosa	4	est 100	Less than 2	M	G	Roadside hedgerow Regularly trimmed to maintain a consolidated form and shape Dead elm specimens present throughout	5	1.2	C (ii)
H33	Hawthorn Crataegus monogyna Blackthorn Prunus spinosa	2	est 100	Less than 2	M	F	Roadside hedgerow Has not been maintained for several years, vertical growth to 6m Dead elm throughout	5	1.2	C (ii)

	Species	Height	Stem Dia.	Crown Radius	Age Class	Overall Condition	Structural Condition	RPA	RPA Radius	BS5837 Cat
H34	Hawthorn Crataegus monogyna Blackthorn Prunus spinosa Elm Ulmus spp	4-8	est 200	Up to 6	M	F	Internal field boundary hedgerow Contains many dead elm as well as some elder and willow Unmanaged with an outgrown form. The hedgerow is not continuous and has a number of gaps	18	2.4	C (ii)
H35	Hawthorn Crataegus monogyna Blackthorn Prunus spinosa	2	est 100	Less than 2	M	F	Internal field boundary hedgerow Regularly trimmed to maintain a consolidated form and shape. It contains a some elm at it's northern end The hedgerow is not continuous and has a number of gaps	5	1.2	C (ii)
H36	Hawthorn Crataegus monogyna Blackthorn Prunus spinosa	2	est 100	Less than 2	M	F	Internal field boundary hedgerow Not maintained for several years, dead elm present Has been regularly trimmed in the past to maintain a consolidated form and shape though is now slightly outgrown. The blackthorn is predominantly at either end of the hedge with the central section being mainly hawthorn. The eastern end contains some elm and the middle section is broken	5	1.2	C (ii)
H37	Hawthorn Crataegus monogyna Blackthorn Prunus spinosa	2	est 100	Less than 2	M	G	Roadside hedgerow Has been regularly trimmed in the past to maintain a consolidated form and shape though is now slightly outgrown. Contains some crab apple and goat willow	5	1.2	C (ii)
H38	Blackthorn Prunus spinosa	4	est 150	Less than 3	M	G	Internal field boundary hedgerow Has been regularly trimmed in the past to maintain a consolidated form and shape though is now slightly outgrown, no recent management	10	1.8	C (ii)
H39	Elm Ulmus spp	Up to 7	est 200	4-6	M	P	Boundary hedge to Red House It is unmanaged and outgrown with dense brambles. There are some dead and broken sections	18	2.4	C (ii)
H40	Hawthorn Crataegus monogyna Blackthorn Prunus spinosa	1.5	est 100	Less than 2	M	G	Roadside hedgerow Regularly trimmed to maintain a consolidated form and shape. Predominantly hawthorn	5	1.2	C (ii)
H41	Field Maple Acer campestre Blackthorn Prunus spinosa Hawthorn Crataegus monogyna	Up to 5	est 100	Less than 2	M	G	Boundary between the site and the Carpenters Arms public house Has been regularly trimmed in the past to maintain a consolidated form and shape though is now slightly outgrown. It has a dense structure No vertical management	5	1.2	C (ii)

	Species	Height	Stem Dia.	Crown Radius	Age Class	Overall Condition	Structural Condition	RPA	RPA Radius	BS5837 Cat
H47	Hawthorn Crataegus monogyna Hazel	6 max. Most 4	est 200	4-6	M	F	Boundary with private fields Unmanaged with an outgrown form. It contains some field maple, crab apple and elm. There are two ash standards at southern end and some ivy	18	2.4	C (ii)
H48	Hawthorn Crataegus monogyna Blackthorn Prunus spinosa Elm Ulmus spp	Up to 6 (Most 3)	est 100	Less than 2	M	G	Roadside hedgerow Unmanaged outgrown and dense form with a high elm content, a number of which are dead	5	1.2	C (ii)
H49	Hawthorn Crataegus monogyna Blackthorn Prunus spinosa Elm Ulmus spp Field Maple Acer campestre Common Ash Fraxinus excelsior	2	est 100	Less than 2	M	G	Roadside hedgerow Regularly trimmed to maintain a consolidated form and shape and includes some field maple and ash. There is dense ivy present	5	1.2	C (ii)
H51	Hawthorn Crataegus monogyna	6	est 250	8	M	F	Boundary with private fields Multi stemmed hawthorn which is unmanaged and has an outgrown form and is broken in parts. There are some dead stems and ivy present	28	3.0	C (ii)
H88	Blackthorn Prunus spinosa Goat Willow Salix caprea Hawthorn Crataegus monogyna	5	est 150	1.5	M	F/P	Scrappy form hedgerow running along the northern edge of the verge to the north of the A420 Historically managed laterally but not for c.3 years No vertical management apparent	10	1.8	C (ii)

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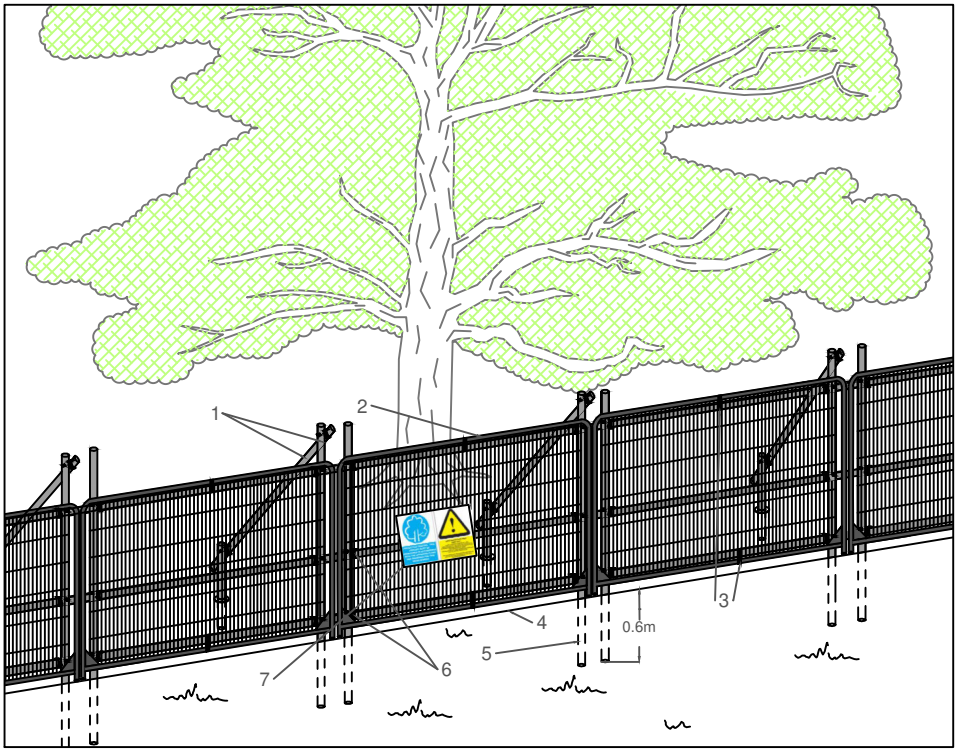
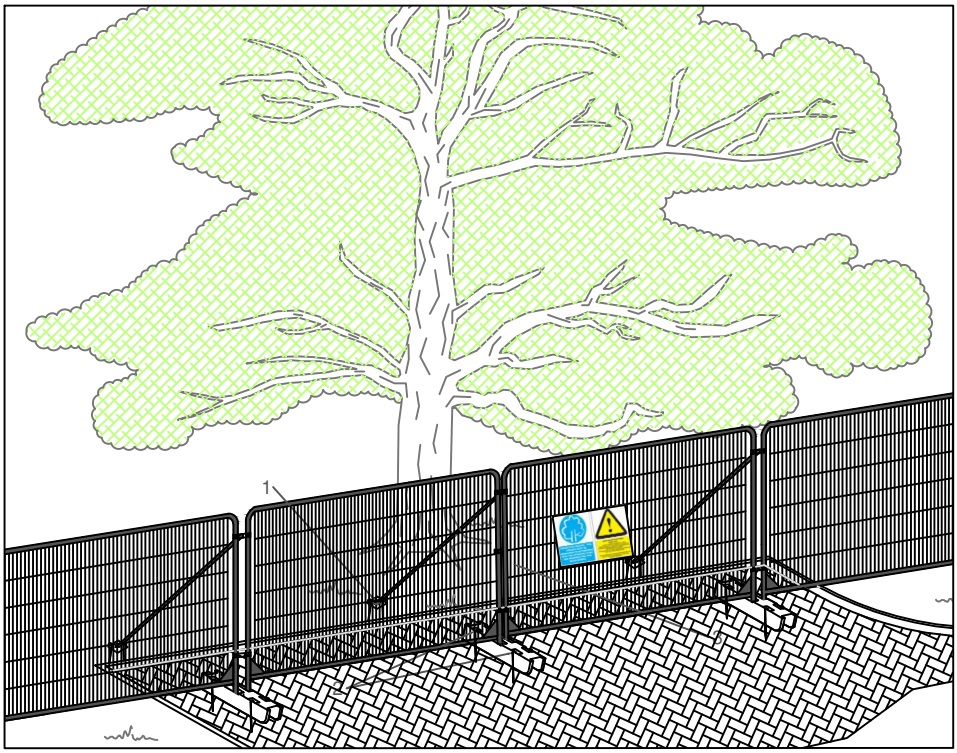


Standard specification for protective barrier

1. Standard scaffold poles
2. Heavy gauge 2m tall galvanized tube and welded mesh infill panels
3. Panels secured to scaffold frame with wire ties
4. Ground level
5. Uprights driven into the ground until secure (min depth of 0.6m)
6. Standard scaffold clamps
7. Construction Exclusion Zone signs



Above ground stabilising systems

1. Stabiliser strut with base plate secured with ground pins
2. Feet blocks secured with ground pins
3. Construction Exclusion Zone signs

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NOTES

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APPENDIX B
PROTECTIVE FENCING SPECIFICATIONS

CAD file: S:\Arb resources\Basic Templates\Tree Protection\Appendix B - Protective Fencing A4.dwg